PERFORMANCE
CONSTRUCTION CHEMICALS

SWADI®
a step ahead...

Micro Concrete - MC07

Flowable High Strength Mortar for New and Repair Damaged Reinforced Concrete

Description

SWADI Micro Concrete - MC07 is a ready to use dry powder which requires only addition of clean water at site to produce a free flowing non shrink repair micro concrete. It is composed of good quality cement, properly selected aggregates & additives. It is used & designed for new and repair of damaged reinforced concrete elements, particularly where access is restricted and where vibration of the placed material is difficult or impossible.

Where to Use

1.) Large volume concrete for new and old repair substrate. 2.) Repair of corrosion damaged concrete. 3.) Jacketing of beams, columns and other structural elements for strengthening. 4.) Pile head re-profiling.

Advantages

- SWADI Micro Concrete - MC07 can be pumped or poured into restricted locations.
- Flowable mortar hence does not require compaction.
- No primer required.
- Very high and ultimate compressive strength.
- Low permeability: provides very good protection against carbon dioxide and chlorides.
- Durability - makes repaired sections highly durable & compatible to parent concrete.
- Rapid strength gain helps in early reinstatement & removal of shuttering.
- Offers excellent resistance to moisture ingress.
- Can be applied at 100 mm thickness at one stroke.

Technical Data

Consistency	Lumps free dry powder (min w/p gives max strength)
Colour	Concrete grey
Yield	Approx 13 litres for 30 kg bag + 4.8 litres

Mix ratio	Ratio	Consistency	Percentage
	Water/ solid ratio	• Flowable	• 0.16%
		• Pourable	• 0.15%

Compressive strength	Flowable (w/p - 0.16%)	Pourable (w/p -0.15%)
	7 days - 50 to 55 N/mm²	7 days - 55 to 60 N/mm²
	28 days - 60 to 65 N/mm²	28 days - 60 to 65 N/mm²
Compressive strength with addition of @10 mm aggregates	w/p - 0.16%	7 days – 55 to 65 N/mm² 28 days – 60 to 70 N/mm²
For aggregate	w/p- 0.18% Aggregate quantity – 0.40%, 10 kg @10 mm sieved, well graded aggregates for 30 kg bag. *Use for >100 mm thicker sections*	
Tensile strength	2 N/mm² @ 28 days	
Flexural strength	6.5 N/mm² @ 28 days	
Wet density	2230 kg/cm³	
Coverage	2.2 kg per 1 sqm per 1 mm thickness	
Storage & shelf life	Shelf life is 12 months from the date of manufacturing. Store at cool & dry place in unopened condition.	
Packaging	30 kg	

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Application Instructions

Surface Preparation

- The surface and remove loose concrete, dust, oil, paint, grease, waterproof coating, etc.
- Any corroded steel in the repair area and remove all scale and corrosion deposits.
- Saturate the substrate with water to prevent absorption of water from the mixed material.
- In case of a very old concrete substrate, it should be primed by applying one coat of SWADI Epoxy Bonding Agent.

Mixing

- Water /powder ratio is 0.16 per 30 kg bag (4.8 litres for 30 kg bag).
- Start addition of SWADI - MC07 powder slowly under continuous mixing. Keep mixing for 2 minutes. Add balance quantity 15 - 20% of mixing water & again mix for another 2 to 3 minutes to form a homogeneous, free flowing, uniform & lump-free mix.
- Aggregates should be used for thickness more than 100 mm with the ratio of 0.40 %(approx 10 kg aggregates per 30 kg SWADI - MC07, Grade as per site conditions.
- For the repairs of damped structures or dampness of reinforced walls, slabs use micro concrete with SWADI SBR (latex) to act as water proofing agent.
- Mixing proportion for damped structures would be (SWADI latex 1 : Water 3 : Micro Concrete 10 : Sand 30) and for further more details refer SWADI SBR (latex) technical data sheet.

Placing

Ensure all free water is drained from the formwork and drain holes are plugged before placing of mixture. The mixed mortar shall be poured continuously to prevent air entrapment. Tapping the formwork with a hammer lightly will ensure effective compaction of the micro-concrete.

Curing

Repaired areas should be thoroughly cured either with water, to attain maximum strength.

Cleaning

These remained mixture on tools and equipment can be removed by washing in clean water immediately after use. Any hardened material will need to be removed mechanically.

Precautions

Avoid contact with eyes and prolonged contact with the skin. Care should be taken to avoid inhalation or ingestion of dust and prevent contact with the eyes. In case of contact with eyes, flush with ample amount of clean water and seek medical advice if necessary.

Condition of Sale

Sold subject to the company's conditions of sale which are available on request.

Storage and Handling Guidelines

The product should be stored in a dry, well-ventilated area, away from moisture, direct sunlight, and children, and in compliance with national regulations. Ensure the packaging is tightly sealed to prevent contamination or moisture ingress, and store it within the recommended temperature range of 5°C to 35°C. Handle the product with care, avoiding direct contact with skin and eyes, and wear appropriate protective equipment such as gloves and masks to prevent inhalation of dust. Use the product as per the instructions provided on the packaging for best results.

Environmental, Health, and Safety Information

For comprehensive information regarding the safe transportation, handling, storage, and disposal of this chemical product, please refer to the accompanying Safety Data Sheet (SDS). The SDS contains detailed information about physical properties, environmental impacts, toxicological hazards, and other safety-related factors. It is crucial to read the current SDS before using any product.

Disclaimer

The information provided in this datasheet is based on our extensive experience and is given in good faith to assist you. However, we reserve the right to update this information at any time without prior notice due to our ongoing research and development efforts.

While we guarantee the consistent quality of our products, we cannot be held responsible for any losses or damages that may arise from their use if site conditions or the execution of work are not properly managed. This data sheet becomes invalid upon the publication of a new edition, and please contact us for the latest version. Please note that we do not provide any warranties regarding the results of our products in

specific applications, as the storage, handling, and application of materials are factors beyond our control.

Contact Information

For immediate orders or enquires, contact our **24/7 support team at +91 8792919513 or enquiry@swadigroup.co.in.** For additional support options and resources, please visit our website at swadigroup.co.in

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